

Empowering Compute, Network, Storage & AI

25G NIC Bonding - Validation & Performance Report

Dual-port 25GbE bonding: Server A (172.16.12.131) <-> Server B (172.16.13.108)

Server / Host	172.16.12.131 <-> 172.16.13.108 (172.16.12.131)
Platform	Server A: AMD EPYC 9124 (251 GiB) / Server B: AMD EPYC 9745 (1.5 TiB) -- back-to-back
Validation Tier	Qualification
Campaign Window	28 May 2026
Prepared For	Netweb Technologies India Limited
Date	2026-05-28

Overall Result: PASS WITH PLATFORM LIMITATIONS

Executive Summary

Validation and benchmark of dual-port 25G NIC performance with and without NIC bonding (link aggregation) between two servers, back-to-back over SFP28 25G-SR fibre (no switch) with jumbo frames. Throughput from iperf3, latency from qperf/ping, across single-link and bonded configurations (balance-rr and balance-xor).

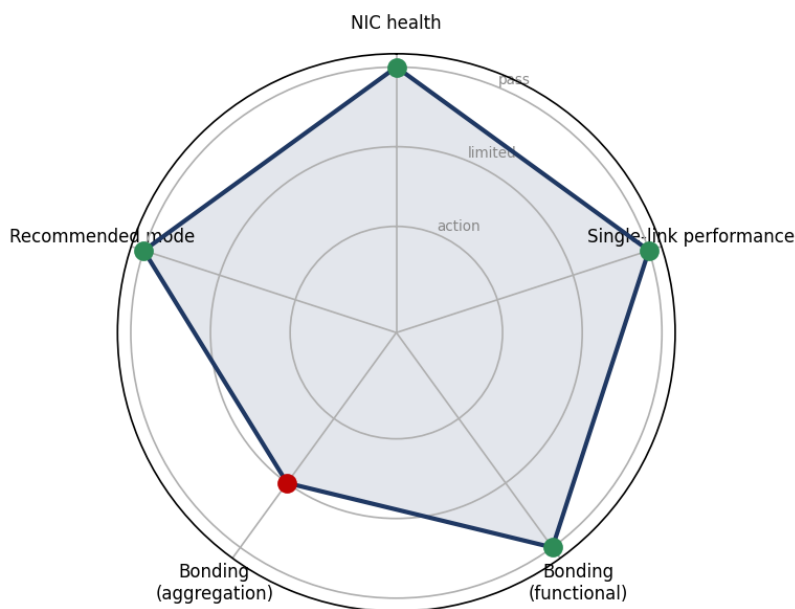
A single 25G link is healthy: ~24 Gb/s (B->A), ~17 Gb/s (A->B), ~18 us latency, near-zero loss with jumbo frames. There is a consistent directional asymmetry (receive-path / CPU contention on the busy Kubernetes node A).

Bonding did not cleanly double throughput - but the cause is the host platform, not the NIC: one of Server A's 25G cards sits in a PCIe x4 slot (31.5 Gb/s) and the only stable two-link bond forced both A ports onto that card, capping aggregate ~31 Gb/s and causing RX loss on B->A; several inter-server fibre links were also marginal (RS-FEC off, mixed optics). balance-rr is unsuitable for TCP (heavy reordering).

Validation Scorecard

Subsystem	Result	Detail	Verdict
NIC health	Both 25G NICs link at 25 Gb/s	Jumbo frames stable, 0% packet loss	PASS
Single-link performance	Up to ~24 Gb/s, ~18 us latency	Near-zero loss with jumbo frames	PASS
Bonding (functional)	Bond forms & carries traffic	Both modes (balance-rr, balance-xor)	PASS
Bonding (aggregation)	~22 Gb/s best (balance-xor, 16 flows)	Capped by host PCIe x4 slot + marginal optics (not a NIC fault)	LIMITED
Recommended mode	balance-xor / 802.3ad (L3+4)	Avoid balance-rr (TCP reordering)	PASS

Subsystem health at a glance



Hardware Inventory

System

Vendor	Tyrone (Netweb)
Model	Two-server back-to-back
Hostname	A 172.16.12.131 / B 172.16.13.108

CPU

Model Server A: AMD EPYC 9124 (16C/64T) | Server B: AMD EPYC 9745 (128C/512T)

Sockets 2

Memory

Total Server A 251 GiB | Server B 1.5 TiB

Type DDR5

Kernel / OS

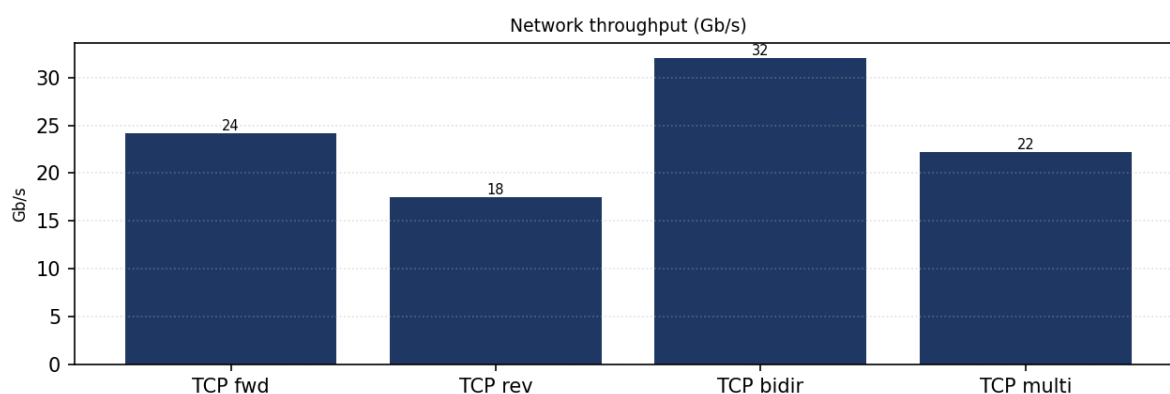
Distro Ubuntu 22.04.4 LTS

Kernel 5.15.0 (A -179-generic, B -94-generic)

Network Interfaces

Interface	Model	Speed	Driver
A NIC1 @ 81:00	Broadcom BCM57414 NetXtreme-E	25 Gb/s (PCIe x4, 31.5 Gb/s slot)	bnxt_en
A NIC2 @ c1:00	Broadcom BCM57414 NetXtreme-E	25 Gb/s (PCIe x8)	bnxt_en
B NIC1 @ 61:00	Broadcom BCM57414 NetXtreme-E	25 Gb/s (PCIe x8)	bnxt_en
B NIC2 @ 91:00	Broadcom BCM57414 NetXtreme-E	25 Gb/s (PCIe x8)	bnxt_en

Network



TCP RX / TX (Gb/s) 24.2 / 17.5

TCP bidirectional (Gb/s) 32.0

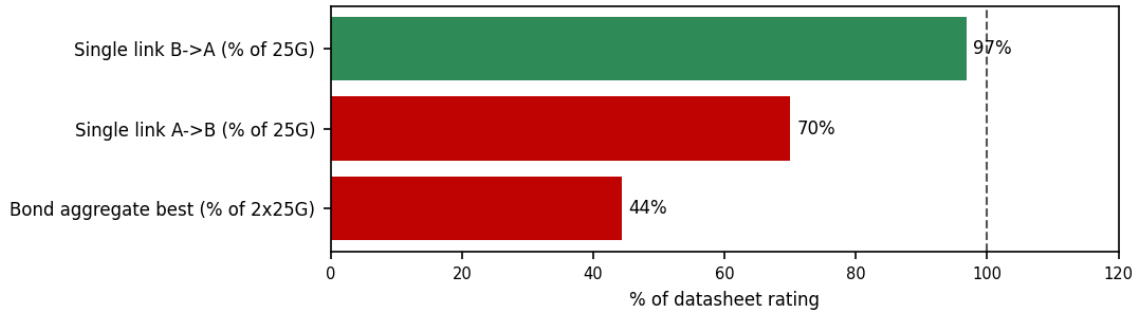
MTU / PFC / ECN 9000 / n/a (back-to-back; RS-FEC off) / n/a

RDMA link n/a (TCP/qperf)

Peer / fabric 172.16.13.108 / direct back-to-back 25G-SR fibre (no switch)

Specification Compliance

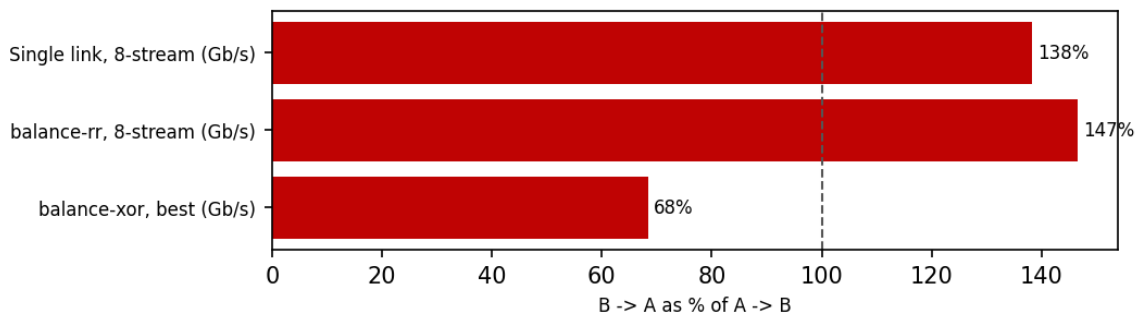
Measured performance as a percentage of datasheet rating. Reference line at 100%. Bars auto-colored: ≥90% green, 75-90% amber, <75% red.



Metric	Measured	Rated	Unit	% of rating
Single link B->A (% of 25G)	24.2	25	Gb/s	97%
Single link A->B (% of 25G)	17.5	25	Gb/s	70%
Bond aggregate best (% of 2x25G)	22.2	50	Gb/s	44%

Cross-Server Comparison

Directional throughput across configurations (iperf3, MTU 9000).



Metric	A -> B	B -> A	Delta
Single link, 8-stream (Gb/s)	17.5	24.2	+38%
balance-rr, 8-stream (Gb/s)	17.4	25.5	+47%
balance-xor, best (Gb/s)	22.2	15.2	-32%

B->A leads on a single link, but bonded B->A degrades because both of Server A's bonded ports share one PCIe x4 slot (RX-side loss). Latency: single-link 18 us (qperf TCP) / 0.45 ms ping; under bond 24 us / 0.52 ms.

Component Health

Kernel log (dmesg) diff: balance-rr produced tens of thousands of TCP retransmissions (packet reordering) -- unsuitable for TCP. Repeated bond reconfiguration stressed the bnxt_en driver and briefly hung Server B's network stack (self-recovered, no reboot).

Findings & Recommendations

Findings

- A single 25G link is healthy: ~24 Gb/s (B->A), ~17 Gb/s (A->B), ~18 us latency, near-zero loss with jumbo frames.
- Directional asymmetry: A->B consistently ~17 Gb/s vs B->A ~24 Gb/s per link (receive-path / CPU contention on the busy Kubernetes node A).

- Bonding did not cleanly double throughput due to host constraints: (a) one of Server A's 25G cards is in a PCIe x4 slot (31.5 Gb/s) and the only stable two-link bond forced both A ports onto it -- capping ~31G and causing RX loss on B->A; (b) several inter-server fibre links were marginal (RS-FEC off, mixed optics).
- balance-rr (round-robin) is unsuitable for TCP -- heavy reordering and tens of thousands of retransmissions with no real gain.
- Repeated bond reconfiguration stressed the bnxt driver and briefly hung Server B's network stack (self-recovered, no reboot).

Recommendations

- Move Server A's 25G card from the PCIe x4 slot into an x8 slot (or bond one port per card across two x8 cards) -- the single biggest blocker to real 2x25G aggregation.
- Use matching 25G-SR optics on both ends of every link, clean/reseat the LC connectors, and enable RS-FEC (ethtool --set-fec encoding rs).
- For production bonding, prefer 802.3ad (LACP) via a switch over back-to-back, and persist the configuration in netplan once links are stable.
- Use balance-xor or 802.3ad (flow hashing) for TCP -- never balance-rr.
- Expected after the above: ~45-48 Gb/s multi-flow aggregate on a 2x25G bond.

Conclusion

Area	Outcome	Verdict
NIC health	Both 25G NICs link at 25 Gb/s; jumbo stable, 0% loss	PASS
Single-link performance	Up to ~24 Gb/s, ~18 us latency	PASS
Bonding (functional)	Bond forms and carries traffic in both modes	PASS
Bonding (aggregation)	Capped by host PCIe x4 slot + marginal optics (not a NIC fault)	LIMITED
Recommended mode	balance-xor / 802.3ad (layer3+4); avoid balance-rr	PASS

Reproducibility Appendix

Exact tooling, kernel command line, BIOS state and a SHA-256 of the raw data bundle, so this run can be audited and reproduced.

Iperf3	iperf3 3.9
Latency Tools	qperf, ping
Ethtool	yes
Distro	Ubuntu 22.04.4 LTS
Kernel	5.15.0 (A -179 / B -94)
Nic	Broadcom BCM57414 NetXtreme-E dual-port 25GbE (bnxt_en)
Optics	SFP28 25G-SR fibre (Cisco-OEM), direct back-to-back (no switch)
Tuning	MTU 9000 (jumbo), RX/TX rings 2047, flow control ON, NUMA-pinned
Campaign Window	28 May 2026
Note	Reconstructed from the validated source report into the CoreBench results.json contract.